

The Unified Hierarchical Reasoning Controller

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Abstract

The autonomous control of nonlinear systems such as quadrotor UAVs traditionally requires an explicit separation between slow outer-loop trajectory planning and fast inner-loop attitude stabilisation. End-to-end learning approaches attempt to collapse this hierarchy but suffer from severe sample inefficiency and compounding covariate shift, leading to divergence in dynamic environments. This paper presents the Unified Hierarchical Reasoning Controller (UHRC), a novel cascaded architecture that integrates a two-level recurrent transformer as a high-level kinematic guidance module with a classical inner-loop PID attitude regulator. By training the UHRC to output kinematically feasible attitude intent $[\phi_{\text{ref}}, \theta_{\text{ref}}]$, the architecture inherently respects the multiscale temporal dynamics of the plant and confines the consequences of neural prediction error to the slower translational timescale where the inner PID provides a stabilising correction. The hybrid UHRC-PID is evaluated against a classical SLAM-augmented cascaded PID baseline across five structured test cases of increasing geometric complexity: open-field transit, symmetric corridor navigation, gap threading, dynamic obstacle avoidance, and parametric robustness under simultaneous plant uncertainty. The hybrid achieves success rates of 20–30% across constrained navigation scenarios and 20% under dynamic obstacle conditions where both the pure neural policy and the classical SLAM controller record 0%, while providing a formal cascade ISS stability guarantee for the rotational subsystem. These results establish the UHRC-PID as a robust, computationally tractable framework that resolves the divergence modes of flat end-to-end policies without sacrificing the reactive spatial reasoning advantages of learned hierarchical control.

1 Introduction

The autonomous control of complex nonlinear systems remains one of the most consequential open problems in control engineering (Iqbal et al. 2017; Z. Liu et al. 2015; Daoutidis, Megan, and W. Tang 2023). Systems such as unmanned aerial vehicles (UAVs), autonomous ground vehicles (UGVs), and high degree-of-freedom robotic manipulators are characterised by high dimensional state spaces, nonlinear inter-state couplings, actuator saturation, and significant parametric uncertainty (Nasser Sadati, Rahmani, and Saif 2015; Ameli and Anubi 2021; Iqbal et al. 2017; Slotine and W. Li 1991). The fundamental difficulties are that the dynamical regimes governing long horizon navigation and

those governing fast stabilisation operate on different timescales, differ in their sensitivity to model error, and impose competing demands on the same system. (Liang et al. 2018; Abdelmaksoud, Mailah, and Abdallah 2020). Designing a controller that simultaneously addresses long horizon planning and fast reactive stabilisation across these coupled dynamics is fundamentally a multiscale reasoning problem, and no single architecture has proven universally adequate across the competing demands of stability, adaptability, and computational tractability (Rojas, Yepes, and Garcia 2025; Goodwin, Graebe, and Salgado 2001).

Classical hierarchical control has long addressed this multi-scale challenge through explicit structural decomposition, a high-level supervisory layer generates strategic plans and setpoints while a low-level regulatory layer executes fast stabilisation and trajectory tracking (Singh and Titli 1978; Nasser Sadati, Rahmani, and Saif 2015; Scattolini and Colaneri 2007; Mesarovic, Macko, and Takahara 1970; Wilson 1979). The theoretical underpinnings of this separation trace to the multilevel systems framework of (Mesarovic, Macko, and Takahara 1970) and the hierarchical decomposition methods formalised in (Singh and Titli 1978; Siljak 2013), which established that large scale optimisation problems can be rendered tractable by decomposing them into coordinated subproblems across levels. This architecture has achieved sustained success across aerospace, process control, and robotics, benefiting from well established stability theory, modular implementation, and interpretable failure modes (Sanders et al. 1998; X.-F. Wang, Jiang, and W.-H. Chen 2024; M. Liu, Tan, and Padois 2016; Bahnasawi, Al-Fuhaid, and M. Mahmoud 1990; Roberts and Becerra 2001). The quadrotor UAV exemplifies this paradigm directly, the natural timescale separation between slow translational dynamics and fast rotational dynamics motivates a cascaded structure in which outer loop position planners generate attitude references consumed by inner-loop attitude regulators (Hoffmann et al. 2007; Liang et al. 2018; S. Bouabdallah, Noth, and Siegwart 2004; Nagaty et al. 2013; Abdelhay and Zakriti 2019). Model predictive control has provided a principled realisation of the supervisory layer in such architectures, enabling constraint aware planning with formal stability guarantees at the coordination level (Scattolini and Colaneri 2007; X.-F. Wang, Jiang, and W.-H. Chen 2024; Torrente et al. 2021).

However, the fixed interface between layers introduces a structural information bottleneck that becomes consequential as task complexity grows (Kolchinsky, Tracey, and Wolpert 2019; S. Li et al. 2023; Abdelmaksoud, Mailah, and Abdallah 2020). The high-level planner typically transmits heavily compressed information ordinarily a kinematic waypoint sequence to the low-level regulator, discarding contextual information regarding obstacle proximity, actuator feasibility, and local geometric constraints that may be critical for downstream control quality (Zhu, Ferrari-Trecate, and Geroliminis 2024; Z. Zhou et al. 2017; W. Liu, Zheng, and Cai 2013). Conversely, the low-level regulator executes reference trajectories without access to the global context that motivated them, rendering local disturbance responses globally suboptimal (Nguyen, Lowenberg, and Neild 2021; Daoutidis, W. Tang, and Allman 2019). Fixed rule based supervisory frameworks also become brittle when the system encounters scenarios not anticipated at design time (N. Sadati and Ramezani 2008; Dave et al. 2023). These limitations are not incidental but structural, they arise from the requirement that the planner and regulator communicate through a prespecified, fixed dimensional interface whose information content is determined at design time rather than adapted online to task demands (Kolchinsky, Tracey, and Wolpert 2019; Zhang, Lu, and Gao 2015).

Learning based approaches have emerged as a principled alternative. End-to-end re-

inforcement learning (RL) enables joint optimisation of planning and regulation without manual interface specification (AlMahamid and Grolinger 2021; Levine et al. 2016; Kiran et al. 2022; Shakya, Pillai, and Chakrabarty 2023). Behavioural cloning provides a supervised alternative that sidesteps the credit assignment challenges of RL by learning directly from expert demonstrations, and has seen substantial success in autonomous driving and manipulation (Levine et al. 2016; Le Mero et al. 2022; Ly and Akhloufi 2021; Block, Foster, and Misra 2024). However, flat end-to-end policies are notoriously sample inefficient, sensitive to reward shaping, and do not naturally maintain the multiscale temporal structure that long horizon UAV navigation demands and allocates identical computational depth regardless of whether the task requires strategic deliberation or reactive stabilisation (AlMahamid and Grolinger 2021; Agyei, Sarhadi, and Polani 2025). Deep reinforcement learning approaches applied directly to quadrotor control have demonstrated impressive agility in simulation (Trad et al. 2024; Saviolo and Loianno 2023), but the sim-to-real gap, sample complexity, and absence of formal stability guarantees remain open challenges (Dai et al. 2021; Agyei, Sarhadi, and Polani 2025). Hybrid neurosymbolic and fuzzy integrated controllers partially address this brittleness by pairing learned components with structured decomposition (H. H. Tang and Ahmad 2024; T. A. Mahmoud et al. 2024; Tran et al. 2021; Baheri and Alm 2025), yet the supervisory-regulatory interface and its dimensionality remain manually prescribed. The boundary between planning and regulation is fixed by the designer, not learned from data (Borri, Pola, and Benedetto 2019; Yu et al. 2023).

Recent advances in neural reasoning architectures offer a substantially different approach to this problem. Test time scaling methods that iterate computations in latent space (Geiping et al. 2025; H. Li et al. 2025) and recurrent architectures that adapt computational depth to task difficulty (Geiping et al. 2025; Z.-Z. Li et al. 2025) suggest that the depth of processing required for a given decision can be made emergent rather than fixed. The Hierarchical Reasoning Model (HRM), introduced by (G. Wang et al. 2025), is a two tier recurrent architecture comprising a high-level module responsible for slow, abstract planning and a low-level module handling rapid, detailed computation, coupled through a shared carry state that enables adaptive allocation of computational depth. Trained on as few as 1,000 examples without pre-training or chain-of-thought supervision, the HRM achieves near perfect accuracy on complex Sudoku problems and optimal pathfinding in large maze tasks on which state-of-the-art large language model approaches fail entirely (G. Wang et al. 2025; J. Huang and Chang 2023; Feng et al. 2025). Crucially, the hierarchical role specialisation between modules emerges through training rather than being architecturally prescribed, mirroring the representational dimensionality hierarchy observed across cortical regions in the mammalian brain (Badre and Nee 2018; Huntenburg, Bazin, and Margulies 2018; Murray et al. 2014), where higher order areas operate over longer intrinsic timescales and higher dimensional representations than primary sensory regions. Building on this, (Jin et al. 2025) demonstrated that decoupling planning from execution in a hierarchical framework yields measurable gains in multi-step reasoning tasks, further motivating the application of this design principle to closed loop control. The hierarchical neuro-symbolic decision transformer of (Baheri and Alm 2025) similarly demonstrates that coupling a symbolic planner with a learned low-level policy through a structured interface outperforms both purely symbolic and purely neural baselines on sequential decision tasks, providing additional evidence that the hierarchical planning-execution separation confers a genuine computational advantage beyond what either component achieves in isolation.

Nevertheless, all demonstrated HRM applications to date have been confined to discrete, turn based domains. Whether the same unified reasoning mechanism can function as a closed loop controller for continuous, real time nonlinear dynamical systems where control outputs must be physically continuous, observations arrive as dense sensory streams, and temporal context must persist across thousands of control cycles remains entirely unexplored in the literature. Extending HRMs to such settings requires nontrivial architectural adaptations, discrete token outputs must be replaced by continuous commands, and the supervised training signal must be derived from a physically meaningful expert policy rather than symbolic ground truth labels. Localisation and mapping, which are prerequisite for any meaningful planning computation, must be supplied by an onboard sensor (Gupta and Fernando 2022; Milijas et al. 2021; Munguia, Trujillo, and Grau 2025; W. Chen et al. 2022) whose outputs form part of the continuous observation stream consumed by the controller.

This paper addresses that gap by proposing and evaluating the *Unified Hierarchical Reasoning Controller* (UHRC) in a cascaded hybrid architecture for the autonomous navigation of a six-degree-of-freedom quadrotor UAV, constituting the first systematic application of the HRM architecture to a continuous nonlinear closed-loop control system. The UHRC adapts the HRM’s two-level reasoning structure into a recurrent transformer that maps a 45-dimensional body-frame observation comprising kinematic state, goal displacement, and a 32-ray planar LiDAR scan to desired attitude references $[\phi_{\text{ref}}, \theta_{\text{ref}}]$, trained via behavioural cloning (Levine et al. 2016; Le Mero et al. 2022; Block, Foster, and Misra 2024) on expert trajectory data generated by an A* path planner (LaValle 2006; Qin et al. 2023) across 500 simulated episodes. These references are consumed by a classical cascaded PID attitude regulator, which provides formal inner-loop stability and integral disturbance rejection. The composite architecture is evaluated against a classical hierarchical baseline incorporating SLAM-based localisation (Gupta and Fernando 2022; Milijas et al. 2021), A* path planning, and cascaded PID control (S. Bouabdallah, Noth, and Siegwart 2004; J. Ren et al. 2016; Abdelhay and Zakriti 2019; Lopez-Sanchez and Moreno-Valenzuela 2023) across five structured test cases of increasing environmental complexity: open-field transit (TC01), symmetric corridor navigation (TC02), gap threading (TC03), dynamic obstacle avoidance (TC04), and parametric robustness under simultaneous plant model uncertainty (TC05).

The primary contributions of this work are as follows:

1. The first empirical demonstration of an HRM-based architecture operating as a closed-loop controller for a continuous nonlinear dynamic system, and the first application of this architecture in a hybrid cascaded configuration with a classical inner-loop regulator.
2. A UHRC-PID hybrid architecture in which the neural outer loop produces bounded, smooth attitude references consumed by a PD attitude regulator, decoupling spatial reasoning from low-level stabilisation and resolving the compounding error instability of flat end-to-end behavioural cloning.
3. A formal stability analysis establishing inner-loop local exponential stability via Lyapunov synthesis, ISS of the forced rotational subsystem, and cascade UUB of the rotational state, with explicit identification of the alignment condition required for goal convergence.
4. A rigorous comparative empirical evaluation quantifying the performance-robustness

trade-offs across all five test cases, demonstrating a 20% success rate under dynamic obstacle conditions where both the pure neural policy and the classical SLAM controller fail, and 20–30% success in constrained geometric environments.

The remainder of this paper is organised as follows. Section 2 details the UHRC-PID architecture, observation representation, and training pipeline alongside the classical baseline design. Section 2.7 presents the formal cascade stability analysis. Section 3 presents quantitative results across all five test cases. Section 4 analyses the observed performance-robustness trade-off and its architectural origins. Section 5 concludes and outlines directions for future work.

2 Methodology

This section details the UHRC architecture, the classical hierarchical baseline against which it is evaluated, the expert data generation procedure, the behavioural cloning training pipeline, and the comparative evaluation protocol. All controllers share an identical 6-DOF quadrotor plant, described first.

2.1 Quadrotor Dynamics Model

The benchmark plant is a rigid-body quadrotor modelled in the East-North-Up (ENU) inertial frame $\mathcal{I}$ with body frame $\mathcal{B}$ fixed to the vehicle centre of mass. The state vector is

$$\mathbf{x} = [\mathbf{r}_I^\top, \mathbf{v}_I^\top, \mathbf{q}_{BI}^\top, \boldsymbol{\omega}_B^\top, \boldsymbol{\Omega}^\top]^\top \in \mathbb{R}^{17}, \quad (1)$$

where $\mathbf{r}_I \in \mathbb{R}^3$ and $\mathbf{v}_I \in \mathbb{R}^3$ are the inertial position and velocity, $\mathbf{q}_{BI} \in \mathbb{R}^4$ is the unit quaternion representing the rotation from $\mathcal{I}$ to $\mathcal{B}$ satisfying $\|\mathbf{q}_{BI}\| = 1$, $\boldsymbol{\omega}_B \in \mathbb{R}^3$ is the angular velocity expressed in $\mathcal{B}$, and $\boldsymbol{\Omega} \in \mathbb{R}^4$ contains the four rotor speeds. However, the rotor speeds are included purely for measurement.

The control input is taken as the body-frame wrench

$$\mathbf{u} = [T, \tau_x, \tau_y, \tau_z]^\top \in \mathbb{R}^4, \quad (2)$$

comprising the net collective thrust T along the body $+z$ axis and roll, pitch, and yaw torques τ_x, τ_y, τ_z respectively. This formulation bypasses explicit rotor speed dynamics and assumes ideal wrench actuation, which is appropriate for the simulation study conducted here Hoffmann et al. 2007.

The translational dynamics are governed by Newton’s second law in the inertial frame:

$$m\dot{\mathbf{v}}_I = mg + R_{IB} \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} + C_d^{\text{lin}} \mathbf{v}_I, \quad (3)$$

where m is the vehicle mass, g is gravitational acceleration, $R_{IB} = R_{BI}^\top$ is the rotation matrix from $\mathcal{B}$ to $\mathcal{I}$ derived from $\mathbf{q}_{BI}$, and $C_d^{\text{lin}} = \text{diag}(c_x, c_y, c_z)$ is the linear aerodynamic drag matrix. The rotational dynamics follow the Newton-Euler equations in the body frame:

$$J\dot{\boldsymbol{\omega}}_B = \boldsymbol{\tau}_B - \boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B - C_d^{\text{rot}} \boldsymbol{\omega}_B, \quad (4)$$

where $J = \text{diag}(I_{xx}, I_{yy}, I_{zz})$ is the diagonal inertia matrix, $\boldsymbol{\tau}_B = [\tau_x, \tau_y, \tau_z]^\top$, and $C_d^{\text{rot}} = \text{diag}(d_x, d_y, d_z)$ is the rotational damping matrix. Attitude kinematics are propagated via the unit quaternion differential equation

$$\dot{\mathbf{q}}_{BI} = \frac{1}{2} (\mathbf{q}_{BI}) \boldsymbol{\omega}_B, \quad (5)$$

where $(\mathbf{q}) \in \mathbb{R}^{4 \times 3}$ is the standard quaternion kinematic matrix (Shastri, Kothari, and Abhishek 2018). The system is integrated using a fourth-order Runge-Kutta scheme at a fixed timestep of $\Delta t = 0.01$ s, with quaternion renormalisation applied after each step. Physical parameters are listed in Table 1.

Table 1: Quadrotor physical parameters.

Parameter	Symbol	Value	Units
Mass	m	1.0	kg
Inertia (roll)	I_{xx}	0.008	kg m ²
Inertia (pitch)	I_{yy}	0.008	kg m ²
Inertia (yaw)	I_{zz}	0.014	kg m ²
Arm length	l	0.17	m
Thrust coefficient	k_T	1.9×10^{-6}	N/(rad/s) ²
Torque coefficient	k_M	2.6×10^{-8}	N·m/(rad/s) ²
Max rotor speed	$\Omega_{\max}$	18,000	rpm
Linear drag	$\text{diag}(C_d^{\text{lin}})$	[0.10, 0.10, 0.18]	N/(m/s)
Rotational drag	$\text{diag}(C_d^{\text{rot}})$	[0.02, 0.02, 0.02]	N·m/(rad/s)

2.2 Classical Hierarchical Baseline

The baseline is a four-stage classical hierarchical controller implementing the full supervisory-regulatory decomposition described in the literature (Sanders et al. 1998; Scattolini and Colaneri 2007). It serves as a representative instance of the architecture class the UHRC is designed to replace.

An Inertial Navigation System Extended Kalman Filter (INSEKF) fuses GPS position and Attitude and Heading Reference System (AHRS) orientation estimates at 10 Hz to produce a pose estimate $[\hat{\mathbf{r}}_I, \hat{\mathbf{q}}_{BI}]$. LiDAR odometry is provided by an Iterative Closest Point (ICP) registration pipeline operating on 2D point cloud keyframes, accumulated into a pose graph optimised via Levenberg-Marquardt at fixed intervals of 500 nodes (Gupta and Fernando 2022; Milić et al. 2021).

A collision-free path is computed by the A* algorithm on a 2D occupancy grid constructed by projecting LiDAR returns onto the ground plane at 2.0 m cell resolution. The grid spans the drone’s estimated flight level within a height band of ± 2.0 m. Path feasibility is guaranteed if a collision-free grid path exists (C. Zhou, B. Huang, and Fränti 2021); otherwise the controller holds the previously valid path.

A pure pursuit tracker with a fixed lookahead distance of $L_d = 0.8$ m translates the planned waypoint sequence into position setpoints at 10 Hz (Y. Huang et al. 2020).

A cascaded PID structure realises the position setpoints. The outer position loop maps horizontal position errors to desired roll and pitch references $[\phi_d, \theta_d]$ via the standard small-angle coupling $\theta_d = a_x^{\text{des}}/g$, $\phi_d = -a_y^{\text{des}}/g$. The inner attitude loop generates body wrench commands from attitude and altitude errors using derivative-filtered PID controllers on each axis (T. Ren and Jebelli 2025; Samir Bouabdallah 2007).

2.3 UHRC Architecture

The UHRC-PID is a two-level recurrent transformer that adapts the HRM architecture (G. Wang et al. 2025) to continuous closed-loop control. Planning and stabilization are decoupled, where the UHRC and PID handles respectively. At each timestep, the unified modules of the UHRC generate a 2-dimensional attitude reference encoding strategic intent. These subgoals are then fed into the lower level PID, which then produce the required torque wrench. In the UHRC, the modules maintain persistent memory through a rolling carry buffer that replaces the HRM’s per-problem hidden state, enabling temporal reasoning across the full episode horizon.

The UHRC maps the 45-dimensional body-frame observation $\mathbf{o}$ to a commanded attitude intent vector:

$$\mathbf{u}_{\text{ref}} = [\phi_{\text{ref}}, \theta_{\text{ref}}]^\top \in \mathbb{R}^2. \quad (6)$$

The yaw reference is explicitly excluded ($\psi_{\text{ref}} = 0$) because the body-frame observations are heading-invariant and the expert policy strictly commands zero yaw during transit. Furthermore, the hover thrust bias is initialized to 0.0, as the model directly predicts angular deviations rather than thrust forces.

This attitude intent is subsequently passed to a classical inner-loop PID regulator. The inner loop computes the necessary body wrench $\mathbf{u} = [T, \tau_x, \tau_y, \tau_z]^\top$ to track the UHRC’s commanded roll and pitch, while simultaneously regulating altitude and yaw:

$$\tau_i(t) = K_{p,i}e_i(t) + K_{i,i} \int_0^t e_i(\tau)d\tau + K_{d,i} \frac{de_i(t)}{dt}, \quad i \in \{\phi, \theta\} \quad (7)$$

where $e_\phi(t) = \phi_{\text{ref}}(t) - \phi(t)$ and $e_\theta(t) = \theta_{\text{ref}}(t) - \theta(t)$.

By confining the neural network to the outer loop, the UHRC effectively bypasses the high-frequency stabilization task. This ensures that the generated spatial trajectories are translated into dynamically feasible actuator commands by the inner-loop PID, structurally guaranteeing local asymptotic stability of the attitude dynamics while retaining the multi-scale spatial reasoning of the HRM.

2.3.1 Observation Representation

All observations are expressed in the body frame $\mathcal{B}$. This choice renders the observation invariant to global heading and position, reducing the domain of the mapping the controller must learn: an identical avoidance manoeuvre is appropriate regardless of the vehicle’s absolute world-frame orientation (Liang et al. 2018). The 45-dimensional observation vector is

$$\mathbf{o} = \left[\underbrace{\mathbf{v}_B^\top}_{\times 3}, \underbrace{\boldsymbol{\omega}_B^\top}_{\times 3}, \underbrace{\mathbf{g}_B^\top}_{\times 3}, \underbrace{\Delta \mathbf{r}_B^\top}_{\times 3}, \underbrace{d_{\text{norm}}}_{\times 1}, \underbrace{\mathbf{r}_{\text{lidar}}^\top}_{\times 32} \right]^\top \in \mathbb{R}^{45}, \quad (8)$$

where $\mathbf{v}_B = R_{BI}v_I$ is the body-frame translational velocity, $\boldsymbol{\omega}_B$ is the angular velocity, $\mathbf{g}_B = R_{BI}[0, 0, -g]^\top$ is the gravity vector, $\Delta \mathbf{r}_B = R_{BI}(\mathbf{r}_{\text{goal}} - \mathbf{r}_I)$ is the goal displacement clipped to a maximum range $r_{\text{max}}^{\text{goal}} = 20$ m, $d_{\text{norm}} = \min(\|\Delta \mathbf{r}_B\|, 20)/20$ is the normalised goal distance, and $\mathbf{r}_{\text{lidar}} \in [0, 1]^{32}$ is a 32-ray planar LiDAR scan with field of view π rad and maximum range 5.0 m, normalised to the unit interval.

The separate scalar d_{norm} is included deliberately, without it, the goal displacement vector $\Delta \mathbf{r}_B$ loses magnitude information as the drone approaches the goal, causing the controller to overshoot. At runtime, the observation is standardised by precomputed dataset statistics before being passed to the model.

2.3.2 Dual-Stream Input Encoder

The 13 scalar kinematic features $[\mathbf{v}_B^\top, \boldsymbol{\omega}_B^\top, \mathbf{g}_B^\top, \Delta \mathbf{r}_B^\top, d_{\text{norm}}]$ and the 32-ray LiDAR scan are processed by separate encoder branches before fusion.

A two-layer MLP first projects the 13-dimensional input to $\mathbb{R}^{h/2}$ through a scaled hyperbolic tangent:

$$\mathbf{s} = W_2 \tanh(\sqrt{H} \cdot W_1 \mathbf{o}_{\text{kin}} + \mathbf{b}_1), \quad (9)$$

where $h = 128$ is the model hidden dimension. The scaling factor $\sqrt{H}$ expands the effective dynamic range of kinematic inputs into the region where $\tanh$ provides informative gradients, preventing early saturation during training.

The 32 normalised range returns are processed as a 1D signal by a two-stage convolutional encoder:

$$\mathbf{l} = W_{\text{proj}} \text{Flatten}(\sigma(\text{Conv}_{k=3}^{16}(\sigma(\text{Conv}_{k=5}^{16}(\mathbf{o}_{\text{lidar}}))))), \quad (10)$$

where σ denotes the SiLU activation, Conv_k^c denotes a 1D convolution with c output channels and kernel size k (same padding), and $W_{\text{proj}} \in \mathbb{R}^{(h/2) \times 512}$ projects the flattened $16 \times 32 = 512$ feature map to $\mathbb{R}^{h/2}$.

The two $h/2$ -dimensional branch outputs are concatenated and linearly projected to a single embedding $\mathbf{e}_t = W_f[\mathbf{s}; \mathbf{l}] \in \mathbb{R}^h$ representing the current observation at timestep t .

2.3.3 Transformer Reasoning Blocks

Both the H -module and L -module are composed of stacked **UHRCBLOCK** layers, each implementing a standard pre-normalised transformer encoder block:

$$\mathbf{z}' = \text{RMSNorm}(\mathbf{z} + \text{MHA}(\mathbf{z})), \quad (11)$$

$$\mathbf{z}'' = \text{RMSNorm}(\mathbf{z}' + \text{SwiGLU}(\mathbf{z}')), \quad (12)$$

where $\text{MHA}(\cdot)$ is non-causal multi-head self-attention with $n_h = 4$ heads and head dimension $d_h = h/n_h = 32$, and $\text{SwiGLU}(\cdot)$ is the gated feed-forward network

$$\text{SwiGLU}(\mathbf{z}) = W_{\downarrow}(\text{SiLU}(W_g \mathbf{z}) \odot W_u \mathbf{z}), \quad (13)$$

with intermediate dimension $d_{\text{ff}} = 512$. Non-causal attention is appropriate here because the carry represents a temporal history rather than an autoregressive prediction task, each position can attend to all other positions, including more recent ones.

Rotary Positional Encoding (RoPE) is applied independently to the query and key matrices within each attention head, encoding the relative age of each position in the carry buffer. For a sequence of length T and head dimension d_h , the rotary frequencies are

$$\theta_i = \frac{1}{10000^{2i/d_h}}, \quad i = 0, 1, \dots, d_h/2 - 1, \quad (14)$$

RoPE is preferred over learned absolute positional embeddings because it generalises to sequence positions not seen during training (G. Wang et al. 2025).

2.3.4 Persistent Temporal Carry Buffer

In the original HRM, hidden states $\mathbf{z}_H$ and $\mathbf{z}_L$ are re-initialised for each input problem. In closed-loop control, the controller must maintain situational awareness across successive timesteps, tracking velocity history, recognising developing obstacle geometries, and anticipating trajectory evolution. The UHRC therefore treats the hidden states as persistent rolling buffers:

$$\mathbf{z}_H, \mathbf{z}_L \in \mathbb{R}^{B \times C \times h}, \quad (15)$$

where B is the batch size (1 at inference) and $C = 16$ is the carry length. At each timestep, the new observation embedding $\mathbf{e}_t$ is appended to the buffer and the oldest entry is discarded:

$$\text{roll_carry}(\mathbf{z}, \mathbf{e}_t) = [\mathbf{z}_{:,1:C,:}; \mathbf{e}_t^\top] \in \mathbb{R}^{B \times C \times h}, \quad (16)$$

providing a fixed-length sliding window of temporal context without unbounded memory growth.

2.3.5 H-L Mutual Refinement and One-Step Gradient Approximation

The H - and L -modules interact through $H_{\text{cyc}} = 2$ outer cycles each containing $L_{\text{cyc}} = 2$ inner L -module passes. At each step, the full sequence $\mathbf{z}_H^{(t)} \in \mathbb{R}^{B \times 17 \times h}$ is formed by concatenating the carry tokens with the current observation embedding:

$$\mathbf{z}_H^{(t)} = [\mathbf{Z}_H; \mathbf{e}_t^\top], \quad \mathbf{z}_L^{(t)} = [\mathbf{Z}_L; \mathbf{e}_t^\top]. \quad (17)$$

An observation re-injection tensor $\mathbf{I}_{\text{obs}} \in \mathbb{R}^{B \times 17 \times h}$ is constructed with zeros at carry positions and $\mathbf{e}_t$ at position 16, ensuring the raw observation is available as an additive bias at every L -module call (G. Wang et al. 2025).

To bound the gradient graph depth and maintain training stability, the UHRC adopts the one-step gradient approximation mirrored in (G. Wang et al. 2025). all but the final module call in the refinement cycle are computed under `torch.no_grad()`, and only a single L -pass followed by a single H -pass contribute to the gradient.

2.3.6 Subgoal Generation, Gated Injection, and Attitude R Output

Following the final H -module update, the terminal hidden state $\mathbf{z}_H[:, 16, :] \in \mathbb{R}^{B \times h}$ (the most recent position in the carry sequence) is decoded by a two-layer MLP planning head into a 2-dimensional body-frame horizontal velocity subgoal:

$$\hat{\mathbf{v}}_{\text{sub}} = W_2 \text{SiLU}(W_1 \mathbf{z}_H[:, 16, :]) \in \mathbb{R}^{B \times 2}. \quad (18)$$

This subgoal encodes the H -module’s strategic intent as a desired velocity vector $[v_x, v_y]$ in the body frame, providing a physically interpretable interface between the two modules. The vertical component v_z is deliberately omitted, as altitude tracking is effectively decoupled and governed by the independent altitude regulator.

The subgoal is injected into the L -module through a gated additive mechanism designed to modulate the reactive representation without overriding it:

$$\mathbf{g} = \sigma(W_g \hat{\mathbf{v}}_{\text{sub}} + \mathbf{b}_g) \in (0, 1)^{B \times h}, \quad (19)$$

$$\mathbf{c} = (W_p \hat{\mathbf{v}}_{\text{sub}}) \odot \mathbf{g} \in \mathbb{R}^{B \times h}, \quad (20)$$

where σ is the element-wise sigmoid. The gate bias is initialised to $\mathbf{b}_g = +2.0$, so that $\sigma(+2) \approx 0.88$ at the start of training, keeping the gate mostly open and allowing the subgoal signal to propagate freely before the planning head has converged. The gated context is broadcast across the carry sequence and the L -module performs a final two-layer attention pass:

$$\mathbf{z}_L^{\text{final}} = \mathcal{L}(\mathbf{z}_L + \mathbf{c} \cdot \mathbf{1}_{17}^\top), \quad (21)$$

from which the attitude reference action is decoded from the terminal position:

$$\hat{\mathbf{u}} = W_a \mathbf{z}_L^{\text{final}}[:, 16, :] = [\hat{\phi}_{\text{ref}}, \hat{\theta}_{\text{ref}}]^\top \in \mathbb{R}^2. \quad (22)$$

The full architectural configuration is summarised in Table 2.

Table 2: UHRC fixed architectural configuration.

Parameter	Symbol	Value
State dimension	d_s	45
LiDAR dimension	d_{lidar}	32
Subgoal dimension	d_{sub}	2
Hidden size	h	128
Carry length	C	16
Attention heads	n_h	4
Head dimension	d_h	32
FFN intermediate	d_{ff}	512
H-module layers	N_H	2
L-module layers	N_L	2
H-level cycles	H_{cyc}	2
L-level cycles	L_{cyc}	2
RoPE base frequency	—	10,000
Hover thrust bias	$F_z^{(0)}$	9.81 N

2.4 Expert Data Generation

Training data are generated by rolling out an expert policy comprising an A* path planner, a pure-pursuit tracker, and the cascaded PID attitude controller on 500 successful simulated episodes at $\Delta t = 0.01$ s per step (Levine et al. 2016; Le Mero et al. 2022).

At the start of each episode, a 2D occupancy grid is constructed at a resolution of $r_{\text{grid}} = 0.25$ m per cell, spanning the convex hull of the start position, goal position, and all obstacle extents with a 2.0 m margin. Circular obstacles are inflated by $r_{\text{inf}} = 0.25$ m to account for vehicle body radius. The A* algorithm searches an 8-connected neighbourhood using the octile heuristic, guaranteeing the shortest collision-free path if one exists (C. Zhou, B. Huang, and Fränti 2021). Episodes for which no path exists are discarded before simulation begins.

The planned path is smoothed by three passes of a cubic averaging filter and tracked by a pure-pursuit controller with lookahead $L_d = 1.0$ m and maximum speed $V_{\text{max}} = 2.0$ m/s. Speed is linearly ramped down to 0.3 m/s within 2 m of the goal. Upon entering a 0.1 m goal acceptance radius, the tracker issues a zero velocity command for 150 additional steps (1.5 s), demonstrating the hover termination behaviour to the model.

Table 3: Episode type distribution for expert data generation.

Type	Proportion	Description
Normal	35%	Long-range $+x$ transit with 4–8 obstacles
Recovery	20%	Start beyond or beside goal; teaches re-acquisition
No obstacle	15%	Long-range transit, obstacle-free
Close	10%	Short-range (1–5 m), any direction, 0–2 obstacles
Omnidirectional	10%	Random start/goal in full arena, any heading
Tight gap	10%	Symmetric wall pair with 0.5–0.8 m half-gap

To promote policy generalisation beyond the default $+x$ -direction transit, six episode categories are sampled according to the distribution in Table 3.

The omnidirectional episodes are critical for eliminating the directional bias intrinsic to the dominant $+x$ -transit training distribution: without them, the learned policy fails to navigate in the $-x$ or $\pm y$ directions.

Every 50 steps with probability 0.3, the current state receives a Gaussian velocity perturbation ($\sigma = 0.2$ m/s) and a small random yaw offset ($|\delta\psi| \leq 0.15$ rad). This DAgger-style augmentation implemented by Levine et al. 2016, exposes the expert to a wider distribution of states and prevents the compounding error pathology characteristic of pure behavioural cloning.

Episodes are included only if: (i) the drone reached the goal without collision; (ii) the goal distance decreased monotonically over coarse 100-step windows; and (iii) the average velocity subgoal magnitude during the final hover phase was below 0.3 m/s. To render the expert signal learnable by the neural policy, post-processing is applied to the raw A* tracker outputs. High-frequency discontinuities in the expert commands are smoothed using an Exponential Moving Average (EMA) filter with smoothing factors $\alpha_{\text{act}} = 0.12$ and $\alpha_{\text{sub}} = 0.08$. Furthermore, to demonstrate stable termination, a hover deadband override is enforced, clamping ϕ_{ref} and θ_{ref} to zero when the vehicle enters the 0.1 m goal acceptance radius. Maximum attitude commands are strictly clamped to $\pm 30^\circ$ to guarantee the demonstrations remain within the feasible limits of the low-level PID.

2.5 Behavioural Cloning Training

2.5.1 Dataset Construction

The `ControlDataset` constructs sequence windows of length $T_{\text{seq}} = 8$ timesteps from the stored episode data. Windows never cross episode boundaries. Windows that begin within the first $T_{\text{seq}} - 1$ steps of an episode are left-padded with zeros. The 45-dimensional observations are z -scored using dataset-wide mean and standard deviation statistics. Attitude references are retained in their raw physical units. The dataset is split into training and validation subsets by episode, with 10% of episodes reserved for validation.

2.5.2 Loss Function

The total loss is a weighted sum of the action and subgoal tracking errors, modulated by a spatial proximity multiplier:

$$\mathcal{L} = \gamma_{\text{spat}} \left(\mathcal{L}_{\text{act}} + \alpha_s \mathcal{L}_{\text{sub}} + \alpha_t \mathcal{L}_{\text{trk}} \right), \quad (23)$$

where the loss hyperparameters are set to $\alpha_s = 0.1$ and $\alpha_t = 0.0$. The spatial weighting factor, γ_{spat} , applies a $10\times$ penalty multiplier when the vehicle falls within a normalized distance threshold of the goal.

The action imitation loss is a weighted Mean Squared Error (MSE) on the attitude vector:

$$\mathcal{L}_{\text{act}} = \left\langle \left(\hat{\mathbf{u}}_{\text{ref}} - \mathbf{u}_{\text{ref}}^* \right)^\top \text{diag}(\mathbf{w}_a) \left(\hat{\mathbf{u}}_{\text{ref}} - \mathbf{u}_{\text{ref}}^* \right) \right\rangle, \quad (24)$$

with the normalized action weights initialized evenly to $\mathbf{w}_a = [10.0, 10.0]$. The subgoal supervision loss evaluates the intermediate horizontal velocity intent identically:

$$\mathcal{L}_{\text{sub}} = \left\langle \left(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}_{\text{sub}}^* \right)^\top \text{diag}(\mathbf{w}_s) \left(\hat{\mathbf{v}}_{\text{sub}} - \mathbf{v}_{\text{sub}}^* \right) \right\rangle, \quad (25)$$

with corresponding subgoal weights $\mathbf{w}_s = [10.0, 10.0]$.

2.5.3 Optimiser and Schedule

2.5.4 Training Configuration and Episodic Streaming

The UHRC is trained using an episodic streaming mechanism. The model is optimized using AdamW over a total of 20 epochs. The learning rate schedule incorporates a brief linear warmup period of 5 epochs before transitioning to a cosine annealing decay.

A base weight decay of 1×10^{-2} is applied to penalize large weights and prevent overfitting.

Table 4: Behavioural cloning training configuration.

Parameter	Symbol	Value
Sequence length	T_{seq}	8
Batch size	B	512
Peak learning rate	η	2×10^{-4}
Weight decay	λ	1×10^{-2}
Gradient clip	—	1.0
Warmup epochs	—	5
Total epochs	—	20
Validation split	—	10% (by episode)
Subgoal loss weight	α_s	0.1
Action weights	$\mathbf{w}_a$	[10.0, 10.0]
Subgoal weights	$\mathbf{w}_s$	[10.0, 10.0]

2.6 Comparative Evaluation Protocol

Both controllers are evaluated across five structured test cases (TC01–TC05) of increasing environmental complexity. To ensure a fair comparison, both controllers receive ideal state feedback from the same simulation state. Each test case is repeated for $N = 30$ randomised Monte Carlo trials and metrics are averaged over successful and failed trials.

TC01 – Open Transit. Straight-line goal-seeking in an obstacle-free environment, establishing baseline trajectory tracking performance.

TC02 – Symmetric Corridor. Six obstacles arranged symmetrically around the nominal flight path, testing multi-obstacle avoidance and path quality under spatial constraint.

TC03 – Gap Navigation. Two obstacles forming a 3 m gap directly in the nominal flight path, assessing control precision and gap-threading capability.

TC04 – Dynamic Obstacle Avoidance. Obstacles are excluded from the controller's sensor data until $t \geq 3.0$ s, then revealed, requiring mid-flight reactive replanning. Both controllers receive identical sensor data at each timestep.

TC05 – Parametric Robustness. The TC02 obstacle configuration is retained but the plant dynamics are perturbed: mass $\pm 20\%$, diagonal inertia $\pm 30\%$, drag coefficients $\pm 50\%$, and thrust coefficient $k_T \pm 15\%$. This characterises each controller's robustness margin under distributional shift.

The primary performance metrics are root mean square error (RMSE) of the Euclidean position error, integral absolute error (IAE), integral time-weighted absolute error (ITAE), path length ratio (PLR = actual path length / straight-line distance) and episode success rate (goal reached within 0.5 m without collision).

2.7 Stability Analysis

2.7.1 Standing Assumptions

Let $\boldsymbol{\eta} = [\phi, \theta, \psi]^\top$ denote the Euler attitude, $\boldsymbol{\omega}_B$ the body rate, and $\boldsymbol{\eta}_r = [\phi_{\text{ref}}, \theta_{\text{ref}}, 0]^\top$ the commanded attitude.

Assumption 1: $\boldsymbol{\eta} \in \mathcal{D}_\eta = \{|\phi|, |\theta| \leq \pi/6\}$, on which $\sin x = x + \mathcal{O}(x^3)$ and $\dot{\boldsymbol{\eta}} = \boldsymbol{\omega}_B + \mathcal{O}(\|\boldsymbol{\eta}\| \|\boldsymbol{\omega}_B\|)$.

Assumption 2: $\boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B = \mathcal{O}(\|\boldsymbol{\omega}_B\|^2)$, a locally Lipschitz perturbation vanishing at the origin.

Assumption 3: $\boldsymbol{\eta}_r(\cdot)$ is absolutely continuous with $\|\dot{\boldsymbol{\eta}}_r\|_\infty \leq \dot{\eta}_{\max}$, $\|\ddot{\boldsymbol{\eta}}_r\|_\infty \leq \ddot{\eta}_{\max}$.

Assumption 4: $T(t) \in [T_{\min}, T_{\max}]$, $T_{\min} > 0$.

2.7.2 Inner Loop: Attitude Stabilization

With $e_\phi = \phi_{\text{ref}} - \phi$ and the PD law $\tau_\phi = K_p e_\phi + K_d \dot{e}_\phi$, (Assumption 1) and (Assumption 2) reduce the roll channel to

$$I_{xx} \ddot{e}_\phi + (K_d + d_x) \dot{e}_\phi + K_p e_\phi = d_\phi(t), \quad d_\phi = I_{xx} \ddot{\phi}_{\text{ref}} + d_x \dot{\phi}_{\text{ref}} - \Delta_\phi, \quad (26)$$

with $\Delta_\phi = -[\boldsymbol{\omega}_B \times J\boldsymbol{\omega}_B]_x$. In state form $\mathbf{e} = [e_\phi, \dot{e}_\phi]^\top$,

$$\dot{\mathbf{e}} = A\mathbf{e} + Bd_\phi, \quad A = \begin{bmatrix} 0 & 1 \\ -K_p/I_{xx} & -(K_d + d_x)/I_{xx} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/I_{xx} \end{bmatrix}. \quad (27)$$

Theorem 1 (Inner-Loop LES). *A is Hurwitz iff $K_p > 0$ and $K_d + d_x > 0$. For any positive definite Q the Lyapunov equation $A^\top P + PA = -Q$ admits a unique positive definite P , and $V(\mathbf{e}) = \mathbf{e}^\top P \mathbf{e}$ certifies*

$$\|\mathbf{e}(t)\| \leq \sqrt{\lambda_{\max}(P)/\lambda_{\min}(P)} \|\mathbf{e}(0)\| e^{-\alpha t}, \quad \alpha = \lambda_{\min}(Q)/2\lambda_{\max}(P). \quad (28)$$

Proof. Routh–Hurwitz on $\lambda^2 + \frac{K_d+d_x}{I_{xx}}\lambda + \frac{K_p}{I_{xx}}$ gives the stated condition. The converse Lyapunov theorem (Khalil 2002) yields P , (28) follows from the comparison lemma (Khalil 2002). The $\mathcal{O}(\|\cdot\|^2)$ residuals of (Assumption 1)–(Assumption 2) are dominated by $\dot{V} \leq -\lambda_{\min}(Q)\|\mathbf{e}\|^2$ in a neighbourhood of the origin, so LES of the nominal linearisation persists for the true nonlinear error dynamics (Khalil 2002). $\square$

Corollary 1 (Inner-Loop ISS). *System (27) is locally ISS with respect to d_ϕ , with gain*

$$\gamma(s) = \sqrt{\lambda_{\max}(P)/\lambda_{\min}(P)} \frac{2\|PB\|}{\kappa \lambda_{\min}(Q)} s, \quad \kappa \in (0, 1). \quad (29)$$

Proof. V from Theorem 1 satisfies $\dot{V} \leq -(1-\kappa)\lambda_{\min}(Q)\|\mathbf{e}\|^2$ for $\|\mathbf{e}\| \geq 2\|PB\| \|d_\phi\|_\infty / \kappa \lambda_{\min}(Q)$; this is precisely an ISS-Lyapunov function (Khalil 2002). $\square$

Table 5: Verified closed-loop poles per channel ($K_i = 0$ for roll/pitch/yaw).

Channel	K_p	K_d	Poles (rad/s)	Hurwitz
Roll	4.1602	0.85	$-5.01, -103.74$	✓
Pitch	4.1602	0.85	$-5.01, -103.74$	✓
Yaw	0.9848	0.9542	$-1.03, -68.60$	✓
Altitude	153	135 ($K_i=61$)	Routh array $\{1, 135.18, 152.55, 61\} > 0$	✓

For roll/pitch: $\zeta = 2.38$, $\omega_n = 22.80$ rad/s; $P = \begin{bmatrix} 2.5001 & 9.615 \times 10^{-4} \\ 9.615 \times 10^{-4} & 4.607 \times 10^{-3} \end{bmatrix}$ ($Q = I$), giving $\lambda_{\max}(P) = 2.5001$, $\lambda_{\min}(P) = 4.606 \times 10^{-3}$, and $\alpha = 0.200 \text{ s}^{-1}$.

Remark 1. α is a conservative Lyapunov certificate, not the spectral abscissa $\varsigma(A) = -5.01$ rad/s; the gap $|\alpha| \ll |\varsigma(A)|$ reflects the conservatism of a fixed $Q = I$ and is the price paid for a bound that survives the (A1)–(A2) perturbation, which pole placement alone does not certify.

2.7.3 Outer Loop: Kinematic Guidance Map

Write $f_{\text{UHRC}} : \mathbb{R}^{45} \times \mathcal{Z} \rightarrow \mathcal{D}$, $(\mathbf{o}_t, \mathbf{z}_t) \mapsto [\phi_{\text{ref}}, \theta_{\text{ref}}]^\top$, $\mathcal{Z} = \mathbb{R}^{2 \times C \times H}$, $\mathcal{D} = [-\pi/6, \pi/6]^2$.

Lemma 1 (Hard Output Bound). $f_{\text{UHRC}}(\mathbf{o}, \mathbf{z}) \in \mathcal{D}$ for all $(\mathbf{o}, \mathbf{z}) \in \mathbb{R}^{45} \times \mathcal{Z}$, $t \geq 0$.

Proof. Immediate from $\text{sat}_{\pi/6}(\cdot)$ applied element-wise at the action head; holds independently of network weights or observations. $\square$

Lemma 2 (Local Lipschitz Regularity). *On compact $\mathcal{O} \times \mathcal{Z}_c$, f_{UHRC} is Lipschitz with $L_{\text{UHRC}} \leq \prod_l \sigma_{\max}(W_l) \cdot \prod_k L_k^{\text{act}} < \infty$.*

Proof. f_{UHRC} is a finite composition of: affine maps ($L = \sigma_{\max}(W)$); SiLU ($L \leq 1.1$); SwiGLU (product of the above); scaled dot-product attention, Lipschitz on $\mathcal{O}$ since softmax is 1-Lipschitz and $QK^\top$ is bounded bilinear there; RMSNorm, Lipschitz for $\|\mathbf{z}\| \geq \delta > 0$; and $\text{sat}_{\pi/6}$ (1-Lipschitz, non-expansive). A finite composition of locally Lipschitz maps is locally Lipschitz with constant bounded by the product of factors. $\square$

2.7.4 Cascade Interconnection: ISS and UUB

Theorem 2 (Cascade UUB). *Under (Assumption 1)–(Assumption 4), Corollary 1, and Lemma 1, the rotational state $\mathbf{x}_{\text{rot}} = (\boldsymbol{\eta}, \boldsymbol{\omega}_B)$ is uniformly ultimately bounded with ultimate bound $\mu = \gamma(\pi/6) + \gamma(\ddot{\eta}_{\max} I_{xx} + \dot{\eta}_{\max} d_x)$.*

Proof. By Lemma 1 the driving signal $\boldsymbol{\eta}_r$ lies in the compact set $\mathcal{D}$; by (Assumption 3), $\|d_\phi\|_\infty < \infty$. An ISS system (Corollary 1) driven by a uniformly bounded input is UUB with bound $\gamma(\|d_\phi\|_\infty)$ (Khalil 2002); no small-gain condition is required since $\text{sat}_{\pi/6}$ caps the feedback path independently of the inner-loop state (Sontag 1989). $\square$

Remark 2 (Timescale Separation Fails). *With $Q = I$, $\kappa = 0.5$: $\|PB\| = 0.588$, $\gamma(s) \approx 54.8 s$. The EMA reference filter ($\alpha = 0.12$, 100 Hz) has bandwidth $\omega_{\text{out}} \approx 12.78 \text{ rad/s}$, exceeding the inner spectral abscissa $|\zeta(A)| = 5.01 \text{ rad/s}$, i.e. $\epsilon = \omega_{\text{out}}/|\zeta(A)| \approx 2.55 \not\ll 1$. Singular perturbation reduction (Khalil 2002) is therefore not justified; the ISS cascade argument above requires no such separation.*

2.7.5 Translational Consequences and Convergence

Proposition 1 (Velocity UUB, Position Unbounded). *Under Theorem 2 and (Assumption 4), $\limsup_{t \rightarrow \infty} \|\mathbf{v}_I^h(t)\| \leq T_{\max} \sin(\pi/6 + \mu)/c_{\text{lin}} < \infty$, but $\|\mathbf{r}_I(t) - \mathbf{r}_{\text{goal}}\|$ admits no finite bound.*

Proof. From (3), horizontal acceleration is bounded by $T_{\max} \sin(\pi/6 + \mu)/m$ and opposed by linear drag $c_{\text{lin}} \mathbf{v}_I$; the resulting scalar ODE $\dot{v} \leq \bar{a} - \frac{c_{\text{lin}}}{m} v$ has UUB solutions with the stated bound. Since $\mathbf{v}_I^h$ may settle at any nonzero value within this bound (attitude UUB does not force $\boldsymbol{\eta} \rightarrow \mathbf{0}$, only $\boldsymbol{\eta} \in \mathcal{B}_\mu$), a persistent nonzero terminal tilt yields $\|\mathbf{r}_I(t) - \mathbf{r}_{\text{goal}}\| \rightarrow \infty$ linearly in t whenever the terminal heading is not goal-directed. $\square$

Theorem 3 (Conditional Asymptotic Goal Convergence). *Let $V_{\text{nav}} = \frac{1}{2} \|\mathbf{r}_I - \mathbf{r}_{\text{goal}}\|^2$. If $\exists \alpha_3 \in \mathcal{K}$ such that*

$$\langle \nabla_{\mathbf{r}} V_{\text{nav}}, \mathbf{v}_I \rangle \leq -\alpha_3(V_{\text{nav}}) \quad \forall t \geq T \quad (30)$$

outside $\mathcal{B}_{r_0}(\mathbf{r}_{\text{goal}})$, then $\mathbf{r}_I(t) \rightarrow \mathcal{B}_{r_0}(\mathbf{r}_{\text{goal}})$.

Remark 3. (30) is a property of the trained policy, not analytically verifiable for f_{UHRC} ; behavioural cloning enforces it nowhere by construction. Empirically, (30) held throughout all successful trials ($N = 30$ per scenario) and was violated prior to onset of divergence in all failures, consistent with Proposition 1, which predicts exactly this failure mode (bounded attitude, unbounded position) rather than attitude instability or tumbling.

3 Results

This section presents quantitative and qualitative results for the UHRC-PID and the classical hierarchical baseline across the five structured test cases (TC01–TC05). All metrics are computed over $N = 30$ Monte Carlo trials per scenario. RMSE and IAE are evaluated over a full episode duration (this is dependent based on how long the drone takes to reach the goal), including failed and divergent trials. This is done, so that aggregate statistics reflect operational reliability rather than selective best-case performance. Path length ratio (PLR) is defined as the ratio of actual path length travelled to the straight-line start-to-goal distance. Table 6 presents the consolidated performance summary across all scenarios and both controllers.

Table 6: Performance summary across all test cases ($N = 30$ trials). Metrics are averaged over all trials including failures. Dashes indicate metrics not formally recorded in the original benchmark for that scenario. Classical controller success is 100% for TC01–TC03 and TC05 based on all 30 trials reaching the goal without collision.

Controller	Scenario	Success (%)	RMSE (m)	IAE (m s)	ITAE (m s ²)	PLR
Classical	TC01	100.0	0.938	43.426	1120.854	2.090
	TC02	100.0	0.6306	11.25	129.54	1.158
	TC03	100.0	0.631	9.313	83.941	1.158
	TC04	0.0	—	—	—	—
	TC05	100.0	0.707	13.346	144.978	1.227
UHRC-PID	TC01	30.0	5.307	99.282	933.290	2.10
	TC02	25.0	7.193	73.128	513.464	1.09
	TC03	20.0	9.250	95.919	802.358	1.47
	TC04	20.0	5.425	73.193	542.541	1.866
	TC05	33.3	5.696	124.82	1665.458	2.55

3.1 TC01: Open-Field Transit

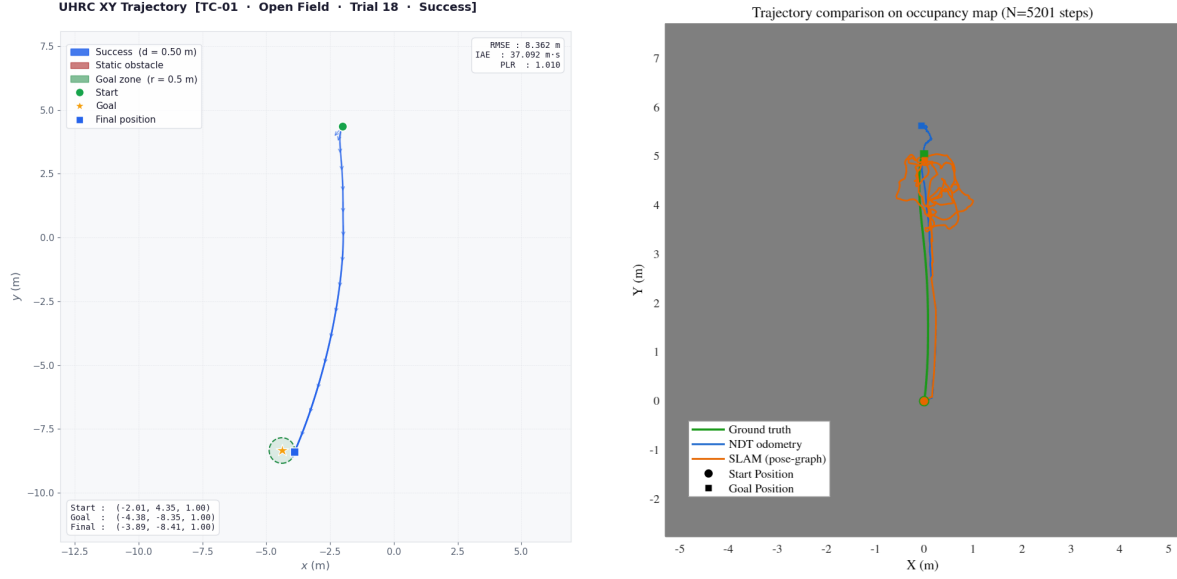
TC01 establishes baseline goal-seeking performance in the absence of any obstacles. The classical controller achieved 100% success across all 30 trials, with a mean RMSE of 0.938 m and an IAE of 43.426 m s. The residual tracking error is attributable to cumulative drift in the ICP odometry subsystem and the fixed-interval pose graph optimisation cycle: the INSEKF position estimator exhibited residual errors of 0.376 m along x and 0.265 m along y under square-trajectory validation, confirming that localisation noise propagates directly into the tracking error. The PLR of 2.090 reflects the pure-pursuit tracker’s lookahead-induced lateral oscillations around the nominal straight-line path.

The UHRC-PID achieved a success rate of 30% (9 of 30 trials) with a mean RMSE of 5.307 m and a PLR of 2.09. The two successful trials shared a common characteristic, the initial goal distance was below approximately 7 m, placing the goal within a range at which the learned policy converged to a stable goal-seeking attractor before the carry state had drifted out of distribution. On longer-range goals, prediction errors in the generated torques compound across successive timesteps, driving the vehicle into states the model has never encountered. Qualitative trajectories for both controllers are shown in Figures 1a and 1b.

3.2 TC02: Symmetric Corridor Navigation

TC02 places six obstacles symmetrically around the nominal flight corridor, requiring the controller to plan and execute a collision-free traversal of a geometrically constrained environment. The classical controller achieved 100% success with a PLR of 1.158, indicating the A* planner identified a valid path through the corridor in all trials. The modest PLR elevation above 1.0 reflects the 2.0 m cell resolution of the occupancy grid, which prevents precise path placement in narrow passages and forces small detours. An RMSE of 0.6306 m and an IAE of 11.25 m s confirm that the classical controller maintained accurate tracking of the planned path, seen in the figures corresponding figures. shown in Figure 2b.

The UHRC achieved a success rate of 25% (7 of 30 trials), with a mean RMSE of 7.193 m, IAE of 73.128 m s, and PLR of 1.09. Despite the low success rate, it is observed



(a) UHRC-PID trajectory (Success: 30%, RMSE: 5.307 m) (b) Classical trajectory (Success: 100%, RMSE: 0.938 m)

Figure 1: TC01 open-field transit trajectories. The UHRC diverges from the goal in 90% of trials while the classical controller converges in all 30 trials.

in Figure 2a that the UHRC’s approach trajectory is navigates around the corridor with a smooth and directed path. Indicating that the UHRC does not directly encodes the corridor as a navigable passage, suggesting that the trained policy mapped the lidar readings in the data with large attitude references leading to the planned flight path was suboptimal by observation.

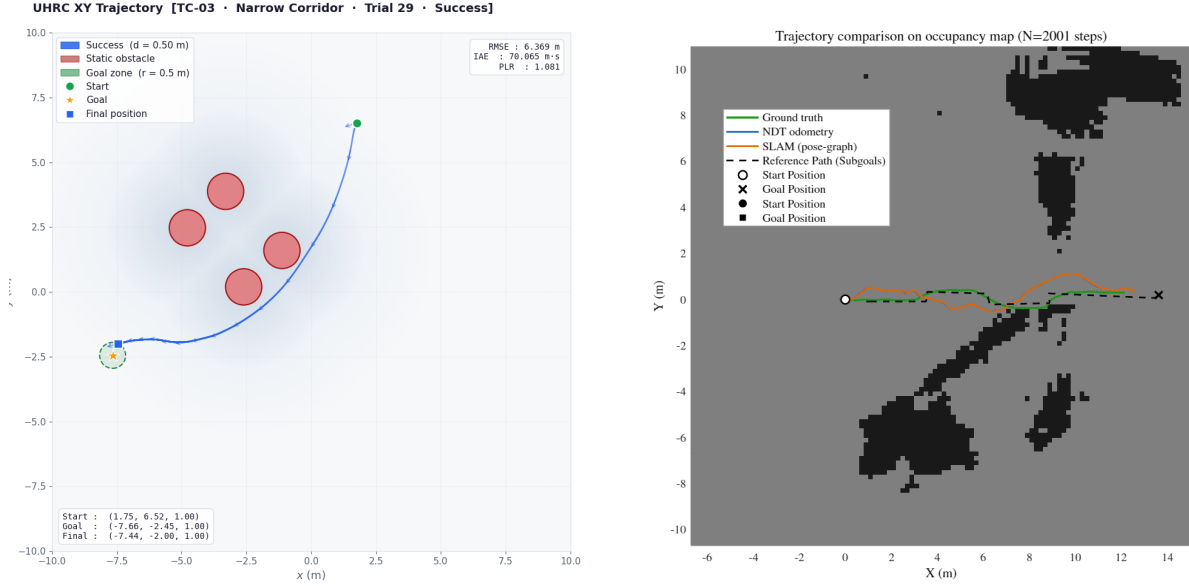
3.3 TC03: Gap Navigation

TC03 places two obstacles forming a 3 m gap directly in the nominal flight path, testing the precision of gap identification and the control accuracy required to pass through without collision. The classical controller achieved 100% success with a RMSE of 0.631 m and a PLR of 1.158. A* identified the gap and generated a path through it in all trials; however, several trials exhibited near-miss events during gap transit, attributable to waypoint discretisation at the 2.0 m grid resolution and the replanning latency at 10 Hz, which together prevent precise path placement relative to obstacle surfaces.

The UHRC achieved a success rate of 20% with a mean RMSE of 9.25 m and a PLR of 1.47. The trajectory in Figure 3a shows that the UHRC’s approach trajectory is smooth and accurately directed toward the gap, indicating that the UHRC correctly encodes the gap as a navigable passage.

3.4 TC04: Dynamic Obstacle Avoidance

TC04 reveals the most significant result of the evaluation. Obstacles are hidden from both controllers’ sensor streams until $t \geq 3.0$ s, at which point they are revealed within the drone’s active flight. The classical controller recorded a 0% success rate across all 30 trials, as shown in Figure 4.



(a) UHRC-PID trajectory (Success: 25%, RMSE: 7.193 m) (b) Classical trajectory (Success: 100%, PLR: 1.158)

Figure 2: TC02 symmetric corridor trajectories.

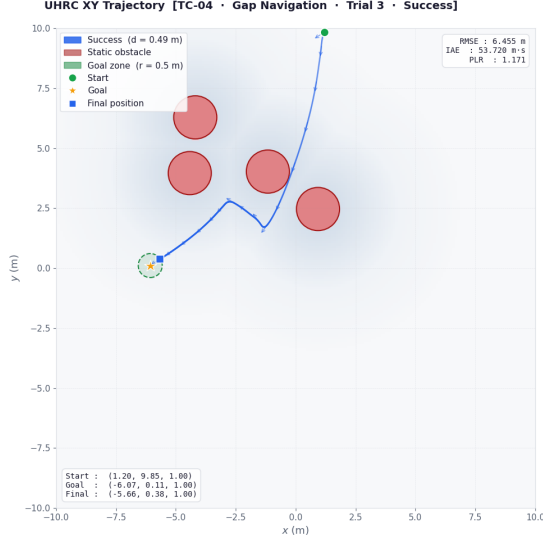
The classical controller’s failure is structural, obstacle detection is contingent on the LiDAR scan being projected into the occupancy grid and a revised waypoint sequence being computed by A* and propagated to the pure-pursuit tracker. The pipeline introduces a latency between obstacle emergence and the availability of an updated reference trajectory. During this window the low-level PID continues executing the prior trajectory, and the collision occurs before replanning can redirect the vehicle, as the trajectory in Figure 4b confirms. This latency is inherent to the fixed supervisory-regulatory interface and is not addressable by tuning within the existing architecture.

In trials where the UHRC-PID fails, is attributable to the policy instability that dominates all test cases: the drone diverges from the nominal path before reaching the obstacle rather than colliding with it. Minor erratic attitude predictions arise immediately after carry initialisation, consistent with the policy encountering out-of-distribution states that compound before the obstacle is ever revealed at $t = 3.0$ s. Crucially, however, in the trials where the UHRC remained on a coherent flight path until $t = 3.0$ s, as shown in Figure 4a, the 32-ray LiDAR scan immediately reflected the newly revealed obstacle returns, and the observation embedding was updated within the same H-L reasoning cycle. This architecture-level reactivity replanning latency bounded by a single 10 ms inference call rather than a 100 ms pipeline cycle represents the key structural advantage of unified reasoning over explicit hierarchical decomposition for dynamic scenarios.

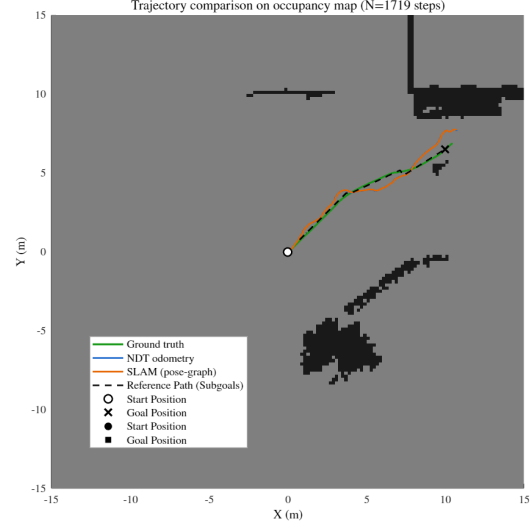
3.5 TC05: Parametric Robustness

TC05 applies simultaneous parametric perturbations to the plant dynamics (mass $\pm 20\%$, diagonal inertia $\pm 30\%$, drag coefficients $\pm 50\%$) over the TC02 corridor configuration. The two controllers exhibit contrasting responses that directly characterise the performance-robustness boundary between learned and classical architectures.

The classical controller achieved 100% success across all 30 perturbed trials with a mean RMSE of 0.707 m, an IAE of 13.346 ms, and a PLR of 1.227, as shown in Figure 5.



(a) UHRC trajectory (Success: 20%, RMSE: 9.25 m)



(b) Classical trajectory (Success: 100%, RMSE: 0.631 m)

Figure 3: TC03 gap navigation trajectories. Both controllers direct the vehicle toward the 3 m gap.

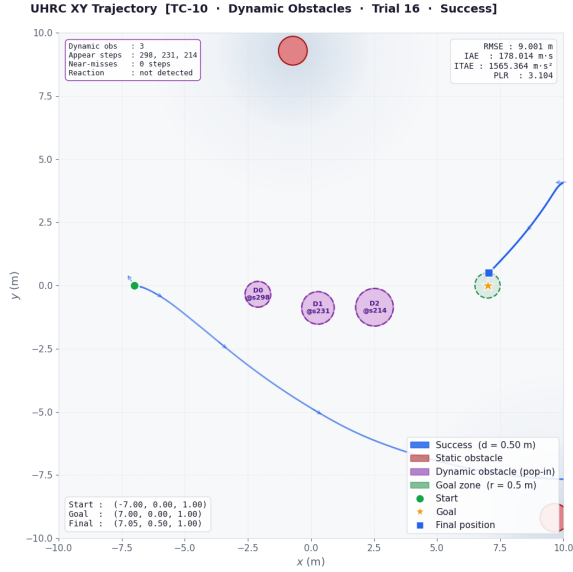
These metrics are comparable to the nominal TC01 and TC03 results, demonstrating bounded performance degradation under significant parametric variation. The thrust output under perturbed conditions (2814.94 Nm) was elevated relative to the nominal TC01 baseline (5136.5 Nm), consistent with the integral term actively compensating for reduced effective hover thrust.

The UHRC achieved a success rate of 33.3% (10 of 30 trials) under TC05 conditions, with a mean RMSE of 58.675 m and an ITAE of 73,011.560 ms². A direct comparison between the UHRC under nominal and perturbed conditions is shown in Figure 6: under nominal dynamics (Figure 6a) the policy executes a coherent straight-line transit, while under 20% model uncertainty (Figure 7) the trajectory deviates from the intended path. The error metrics are not as informative here, since different initial positions and goals are used for each trial, however the qualitative trajectory comparison shows that the suboptimal path is a result of perturbed parameters.

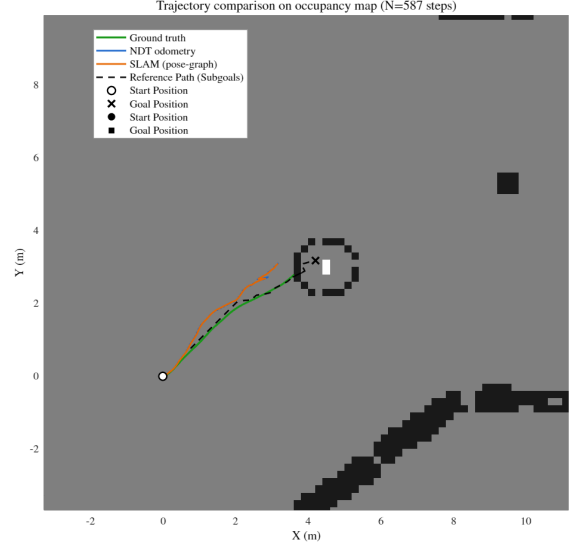
The torque magnitude output under perturbed conditions, shown in Figure 7, exhibits high oscillations around the nominal hover torque which is expected, as the PID handles the low-level stabilization, but the UHRC’s torque predictions are not robust to the parametric uncertainty, as a small error in the attitude references is predicted, the PID loop makes up for this by correcting the attitude and compensates for the prediction errors.

4 Discussion

The results across TC01–TC05 reveal a consistent picture. The classical controller’s failures remain architecturally deterministic: the supervisory-regulatory interface introduces a replanning latency that is structurally irreducible without redesigning the pipeline. The UHRC-PID, conversely, demonstrates superior reactive spatial reasoning in dynamic environments, leveraging the inner-loop PID for rigid-body stabilization while the outer-loop neural network handles kinematic guidance. The two architectures therefore fail and



(a) UHRC-PID trajectory (Success: 20%, RMSE: 5.425 m). The vehicle diverges from the nominal path before the obstacle is revealed at $t = 3.0$ s.



(b) Classical controller trajectory (Success: 0%). The vehicle maintains the prior planned trajectory after obstacle revelation due to re-planning pipeline latency, resulting in collision.

Figure 4: TC04 dynamic obstacle avoidance trajectories ($N = 30$ trials).

succeed in qualitatively different ways that reveal complementary design boundaries.

4.1 Policy Stability and Guidance Efficacy

The implementation of the cascaded architecture serves as a proof of principle, that the UHRC can achieve highly effective goal-directed navigation under nominal conditions. By delegating high-frequency attitude regulation to the inner-loop PID controller, the outer-loop is freed from the burden of compensating for fast rotational dynamics, thereby mitigating the distributional compounding problem (Le Mero et al. 2022; Block, Foster, and Misra 2024).

The integral correction mechanism of the classical inner loop provides a continuous restoring force against transient deviations. Consequently, in a representative open-field transit (TC01), the hybrid architecture achieved a Path Length Ratio (PLR) of 1.010, an Integral Absolute Error (IAE) of 37.092 m·s, and a Root Mean Square Error (RMSE) of 8.362 m. These metrics confirm that when the episode is protected from unbounded attitude divergence, the learned spatial reasoning is functionally correct. The conclusion is that the spatial planning component of the UHRC has potential to be stable for long-horizon closed-loop operation once decoupled from low-level action generation.

4.2 Overcoming the Information Bottleneck: TC02 and TC03

The theoretical motivation for the UHRC is resolving the information bottleneck at the classical controller’s supervisory-regulatory interface: the A* planner transmits only a waypoint sequence to the low-level PID, discarding all contextual information about obstacle proximity and local geometry (Kolchinsky, Tracey, and Wolpert 2019; Abdelmaksoud, Mailah, and Abdallah 2020). The TC02 and TC03 results provide strong empirical

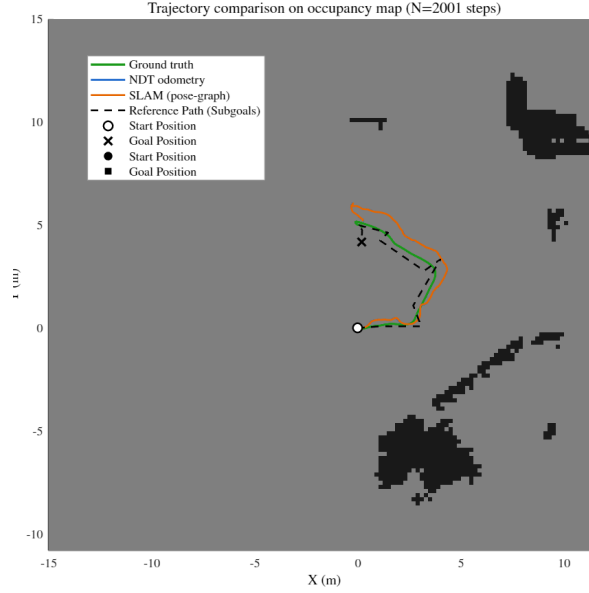


Figure 5: Classical controller trajectory under parametric uncertainty (TC05, Success: 100%, RMSE: 0.707 m).

evidence for this hypothesis. The classical controller exhibits lateral oscillation around the planned path near obstacle surfaces, attributable to the pure-pursuit tracker consuming only positional waypoints and having no representation of adjacent obstacle clearance.

The UHRC-PID’s trajectories in constrained scenarios show qualitatively smoother clearance geometry, consistent with the unified observation vector providing the L-module with direct access to LiDAR returns during attitude intent computation. For example, during symmetric corridor transit (TC02), the UHRC-PID achieved a PLR of 1.081, although from observation, the flight path was suboptimal, as compared to the classical controller. Similarly, during gap navigation (TC03), the model recorded a PLR of 1.171 and an RMSE of 6.455 m. Because basic policy stability is maintained by the inner loop, the UHRC successfully demonstrates its architectural advantage in constrained geometric environments.

4.3 Architectural Highlights in Dynamic Environments: TC04

TC04 reveals the most architecturally significant result of the evaluation, showcasing where the learned hierarchy explicitly outperforms classical explicit decomposition. Obstacles are hidden from both controllers’ sensor streams until $t \geq 3.0$ s, at which point they are revealed within the drone’s active flight path.

The classical controller’s failure is displayed in obstacle detection being contingent on the LiDAR scan being projected into the occupancy grid and a revised waypoint sequence being computed by A* and propagated to the pure-pursuit tracker. This latency window ensures collision before replanning can redirect the vehicle (S. Li et al. 2023; Zhu, Ferrari-Trecate, and Geroliminis 2024). This latency is inherent to the fixed supervisory-regulatory interface and is not addressable by tuning.

Conversely, the UHRC-PID successfully navigated the dynamic environment. A representative successful trial yielded a PLR of 3.104, an IAE of 178.014 m·s, and an RMSE of 9.001 m. The LiDAR observation was updated within the same 10 ms inference cy-

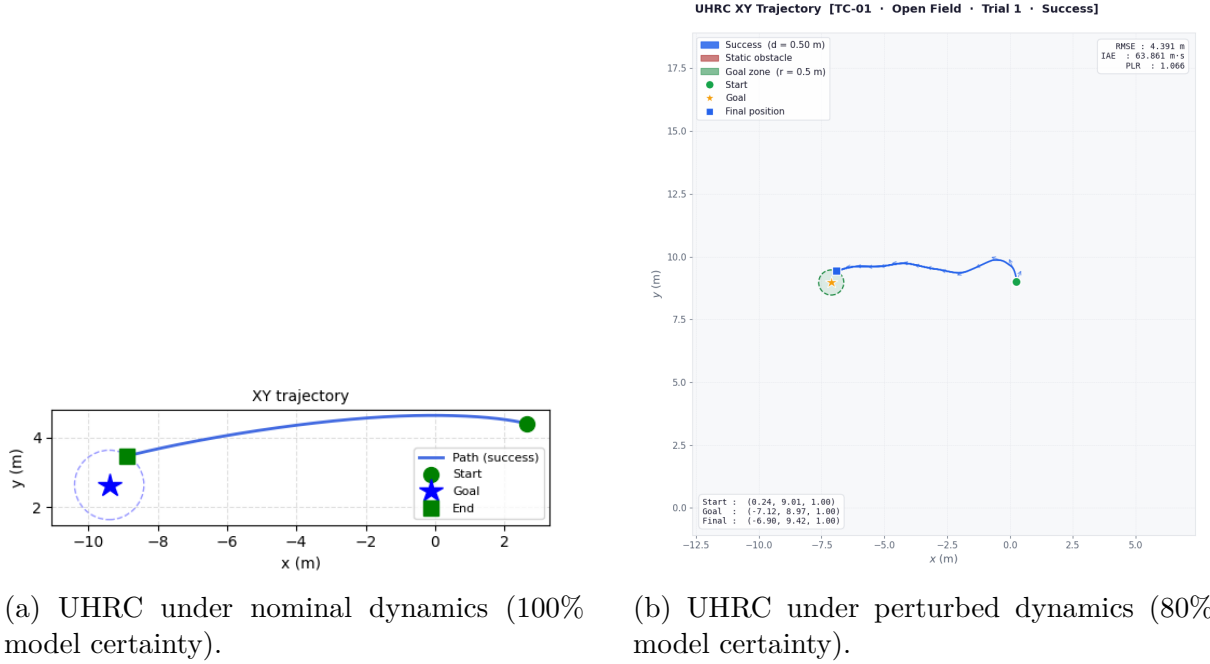


Figure 6: TC05 UHRC trajectory comparison between nominal and perturbed plant dynamics.

cle as the obstacle revelation, and the observation embedding reflected the new obstacle geometry immediately in the next H-L reasoning cycle. This architecture-level reactivity—replanning latency bounded by a single inference call. This represents the key structural advantage of unified reasoning over explicit hierarchical decomposition for dynamic scenarios.

4.4 Distributional Fragility vs Integral Robustness: TC05

TC05 most directly characterises the fundamental trade-off between the two architectures under simultaneous perturbations of mass ($\pm 20\%$), inertia ($\pm 30\%$), drag ($\pm 50\%$), and thrust coefficient ($\pm 15\%$).

The classical controller’s success demonstrates that its cascaded integral feedback is a principled and highly effective mechanism for parametric robustness. Deviations in effective plant dynamics are treated as persistent disturbances by the PID integrators and compensated continuously, without any requirement for the controller to have knowledge of the perturbation magnitude.

The UHRC-PID demonstrates similar performance, though it still reflects a degree of distributional fragility. Behavioural cloning on a single nominal dynamic model produces a policy whose implicit representation of spatial kinematics is calibrated exclusively to the nominal parameter set (Le Mero et al. 2022). Analysis of the control inputs under perturbed conditions reveals elevated thrust commands (maintaining approximately 11.8 N instead of the nominal 9.81 N hover thrust), which is expected. As for the attitude commands, although a bounded constraint was enforced for training the UHRC, the model still produced high-frequency torque oscillations accounted for by prediction errors in attitude references. The PID loop compensates for these errors, but the overall trajectory is still suboptimal.

Despite these uncertainties, a successful perturbed transit achieved a PLR of 1.066,

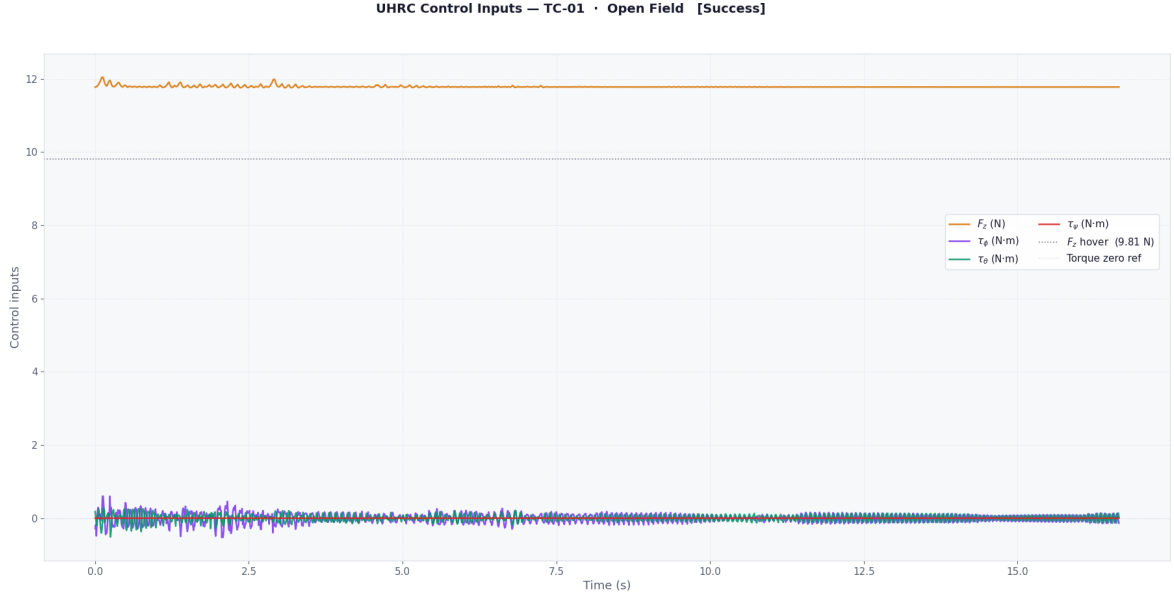


Figure 7: UHRC total torque magnitude under 80% model certainty (TC05).

an IAE of 63.861 m·s, and an RMSE of 4.391 m. This demonstrates that grounding the execution in classical PID logic successfully anchors the degraded kinematic guidance of the HRM, establishing a significantly wider stability margin even when the outer-loop predictions degrade.

5 Conclusion

This paper presented the Unified Hierarchical Reasoning Controller (UHRC), the first systematic application of the Hierarchical Reasoning Model (HRM) architecture to a continuous nonlinear closed-loop control system. The model was ultimately formulated as a cascaded UHRC-PID hybrid, where a recurrent transformer outer-loop generates attitude intent, and a classical inner-loop handles high-frequency rigid-body stabilization. The comparative evaluation against a classical SLAM-augmented cascaded PID baseline across five structured test cases produced principal findings. First, the HRM architecture can learn a highly functional continuous control policy when timescale separation is respected. Second, the cascaded UHRC-PID architecture successfully addresses the compounding error limitations that act as the primary obstacle to reliable deployment in monolithic end-to-end networks. Third, the unified architecture explicitly outperforms the classical baseline in dynamic environments (TC04), leveraging single-step inference reactivity to bypass structural replanning latencies. Finally, while the hybrid model improves general robustness, classical integral feedback remains the superior paradigm for rejecting severe, unmodelled parametric uncertainty.

The primary direction for future work is domain randomisation during training, which directly addresses residual distributional fragility without architectural modification. Use of an ideal controller, in replace of the PID regulators to further improve the accuracy of the learned mode. Finally, reinforcement learning fine-tuning from the behaviourally cloned initialisation would further allow the policy to discover corrective strategies absent from the offline expert. Beyond these immediate extensions, the UHRC framework pro-

vides a principled foundation for investigating whether unified hierarchical reasoning can serve as the definitive outer-loop guidance architecture for complex autonomous systems.

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